Bài toán phân cụm phân cấp theo chiến lược hợp nhất (agglomerative)

1. cài cắm thư viện

In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.cluster import AgglomerativeClustering
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
import scipy.cluster.hierarchy as shc
```

2. tiền xử lí dữ liệu

In [4]:
```
data = pd.read_csv(r'Z:\DOCUMENTS\KPDL\DataMining-Homework\Lab6-Phan-cum\Iris.cs
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

In [5]:
```
# Kiểm tra missing value
data.isnull().sum()
```

Out[5]:
```
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

In [6]:
```
# Xóa cột Id và Species
data = data.drop(['Id', 'Species'], axis = 1)
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 4 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   SepalLengthCm  150 non-null    float64
 1   SepalWidthCm   150 non-null    float64
 2   PetalLengthCm  150 non-null    float64
 3   PetalWidthCm   150 non-null    float64
dtypes: float64(4)
memory usage: 4.8 KB
```

In [7]:
```
# Chuẩn hóa dữ liệu (Normalization)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(data)
X_scaled = pd.DataFrame(X_scaled)
X_scaled
```

Out[7]:

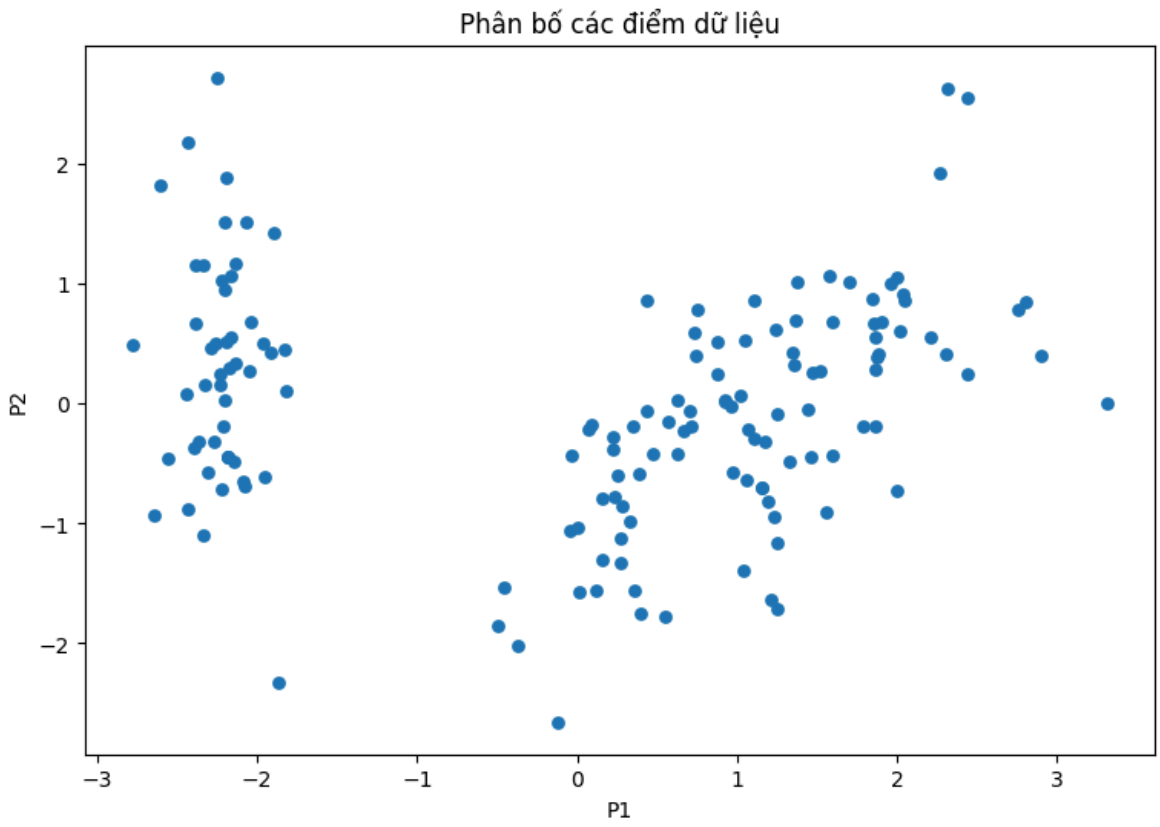
	0	1	2	3
0	-0.900681	1.032057	-1.341272	-1.312977
1	-1.143017	-0.124958	-1.341272	-1.312977
2	-1.385353	0.337848	-1.398138	-1.312977
3	-1.506521	0.106445	-1.284407	-1.312977
4	-1.021849	1.263460	-1.341272	-1.312977
...	...	...	...	...
145	1.038005	-0.124958	0.819624	1.447956
146	0.553333	-1.281972	0.705893	0.922064
147	0.795669	-0.124958	0.819624	1.053537
148	0.432165	0.800654	0.933356	1.447956
149	0.068662	-0.124958	0.762759	0.790591

150 rows × 4 columns

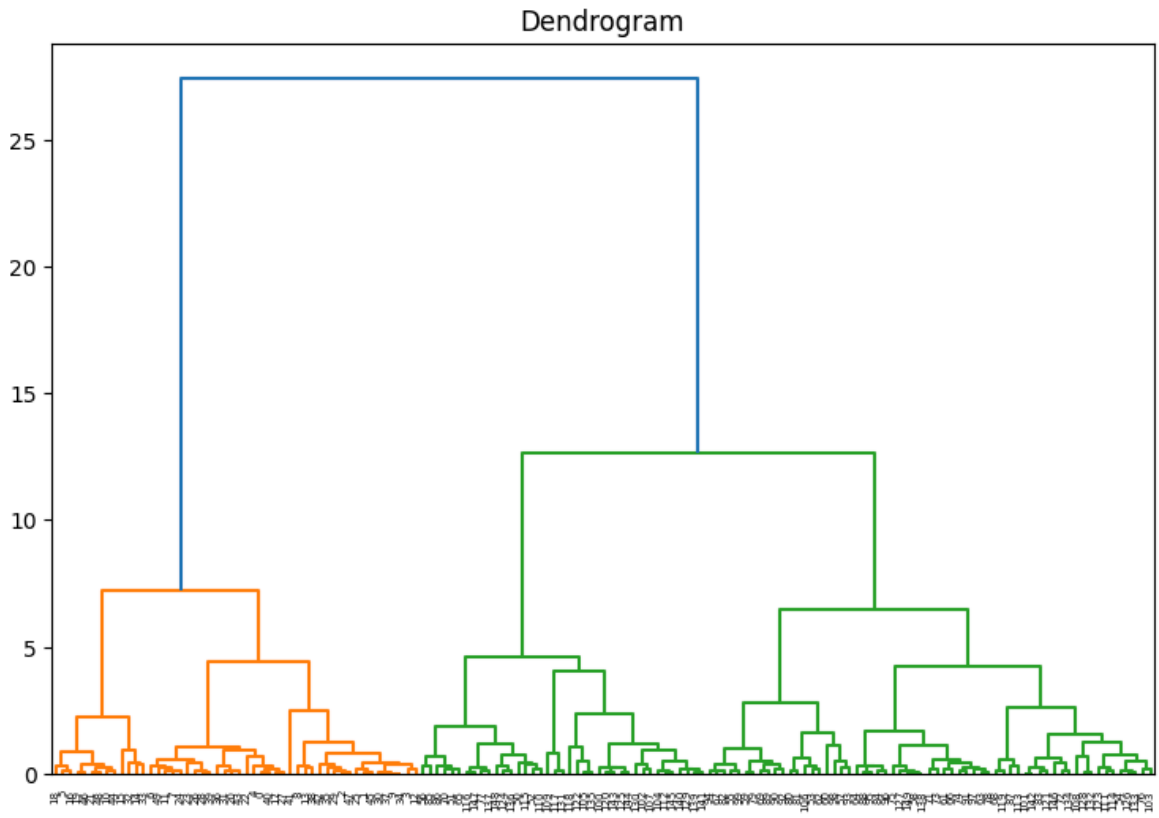
In [8]:
```
# Giảm chiều dữ liệu bằng PCA (Principal component analysis) với n_components =
pca = PCA(n_components = 2)
X_principal = pca.fit_transform(X_scaled)
X_principal = pd.DataFrame(X_principal)
X_principal.columns = ['P1', 'P2']
```

In [9]:
```
# Trực quan hóa các điểm dữ liệu trên đồ thị scatter plot
plt.figure(figsize=(9, 6))
plt.scatter(X_principal['P1'], X_principal['P2'], lw=0, s=40)
plt.xlabel('P1')
plt.ylabel('P2')
plt.title('Phân bố các điểm dữ liệu')
```

Out[9]:
```
Text(0.5, 1.0, 'Phân bố các điểm dữ liệu')
```

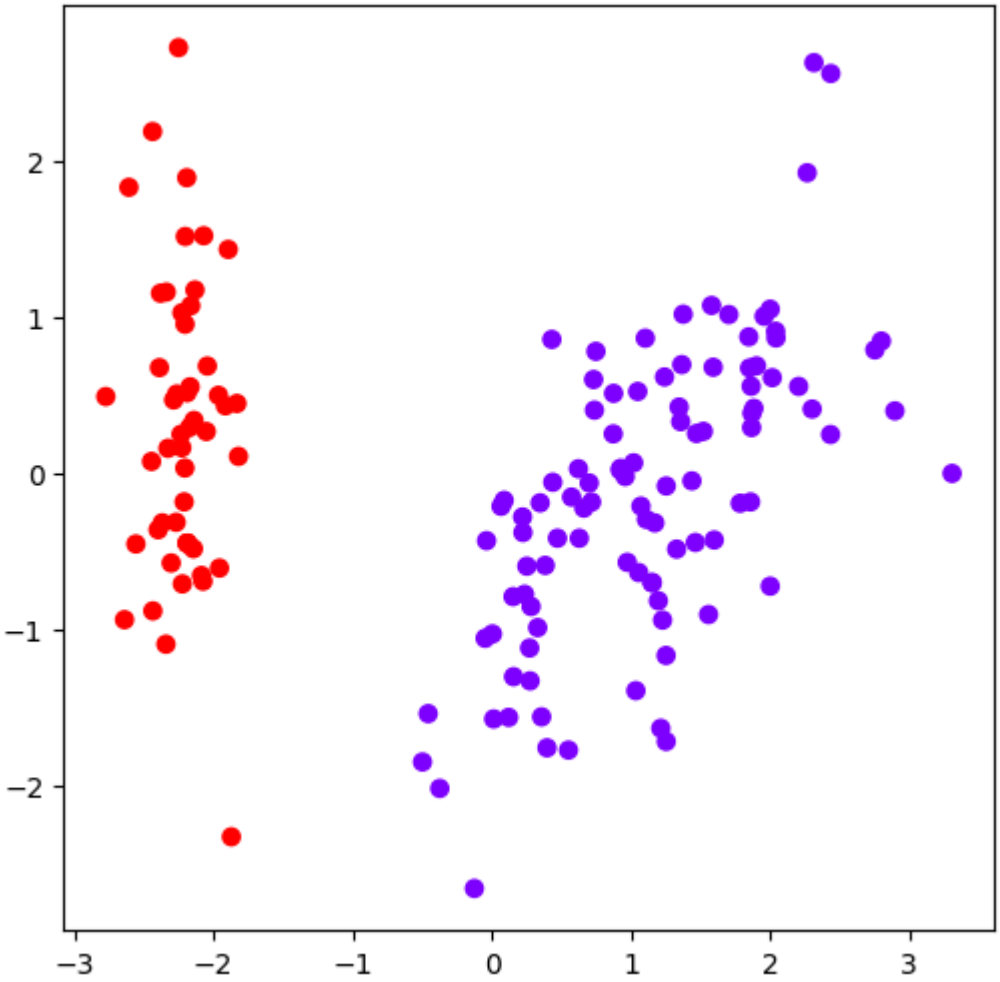


In [10]:
```
plt.figure(figsize =(9, 6))
plt.title('Dendrogram')
Dendrogram = shc.dendrogram((shc.linkage(X_principal, method ='ward')))
```



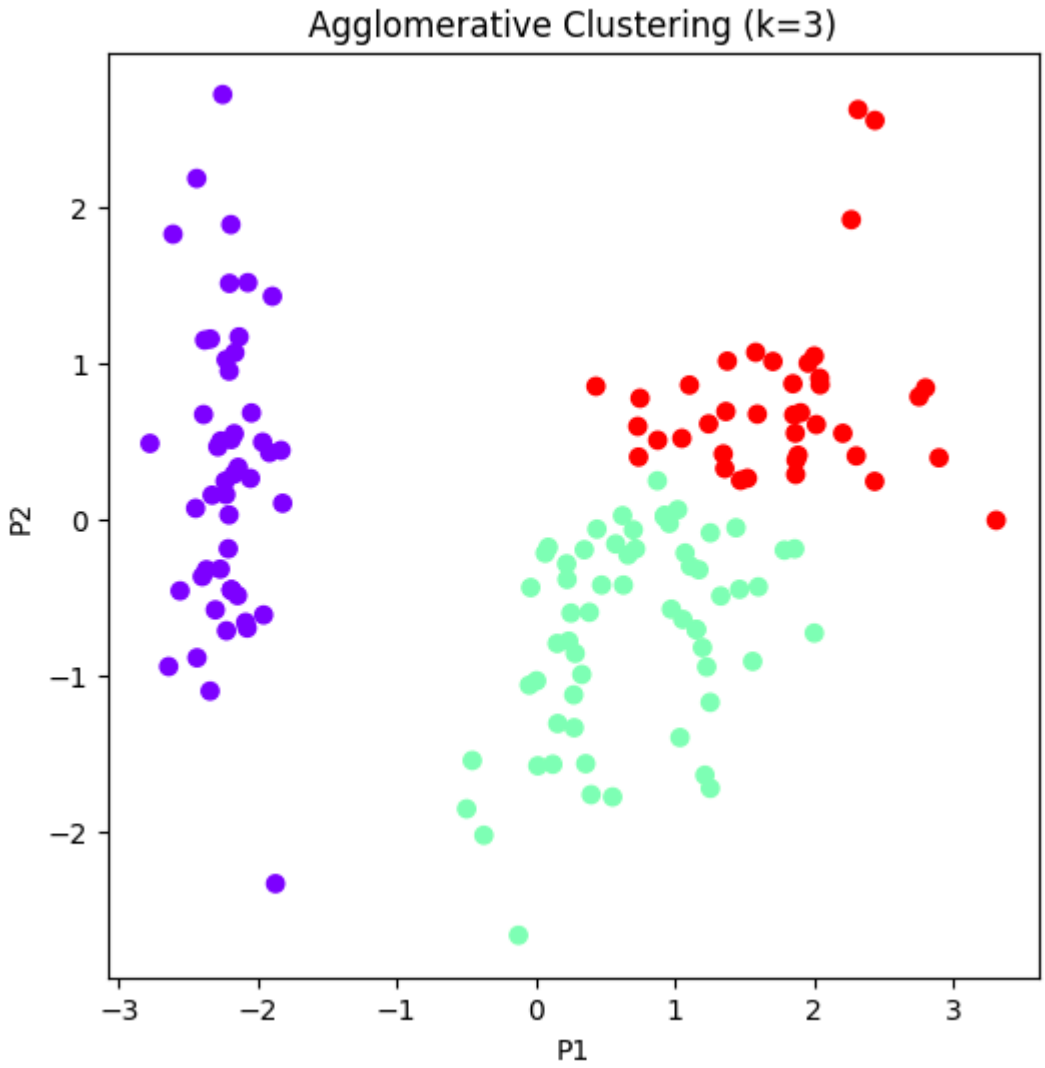
In [12]:
```
ac2 = AgglomerativeClustering(n_clusters = 2)
labels_ac2 = ac2.fit_predict(X_principal)
# Trực quan hóa kết quả phân cụm
plt.figure(figsize =(6, 6))
```

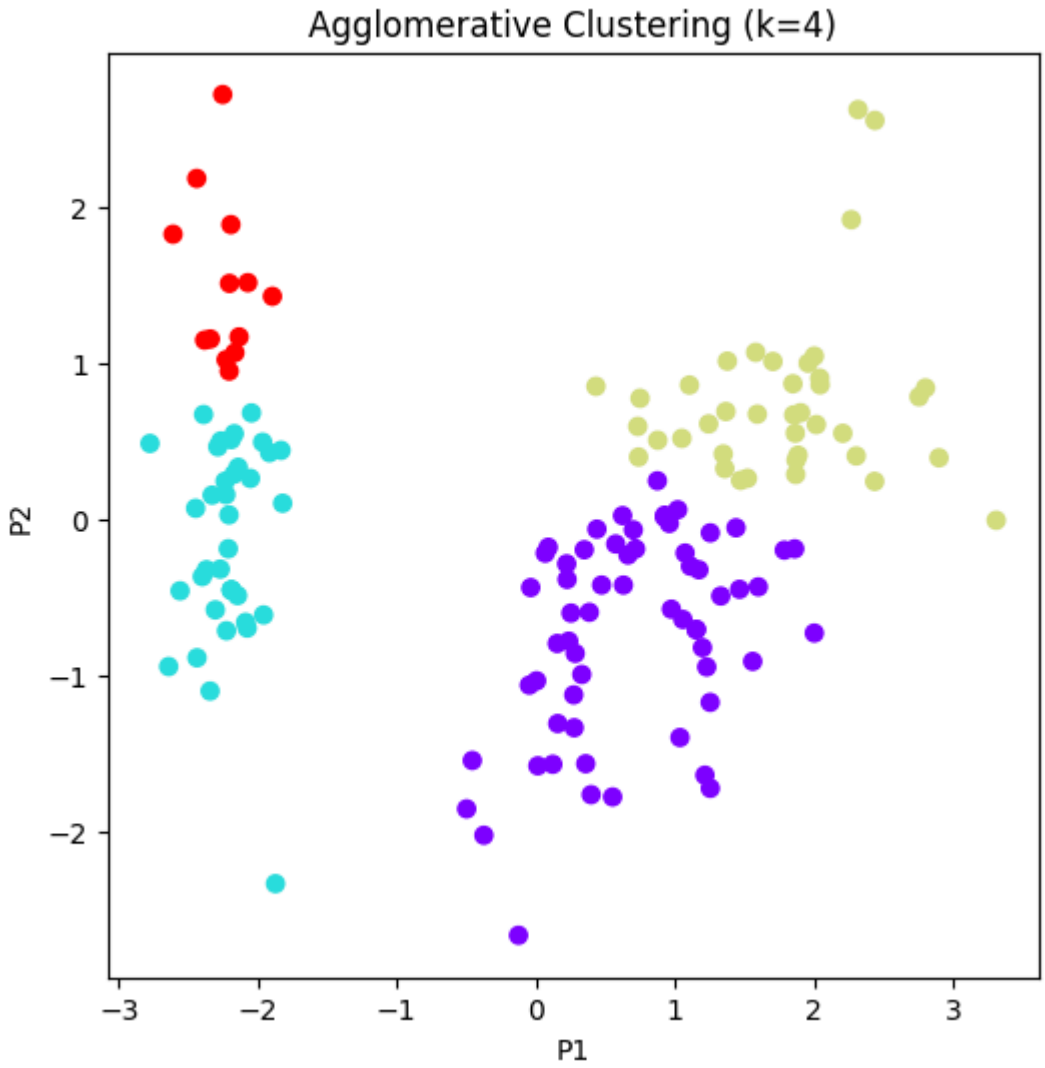
```
plt.scatter(X_principal['P1'], X_principal['P2'],  c = labels_ac2, cmap ='rainbo
plt.show()
```

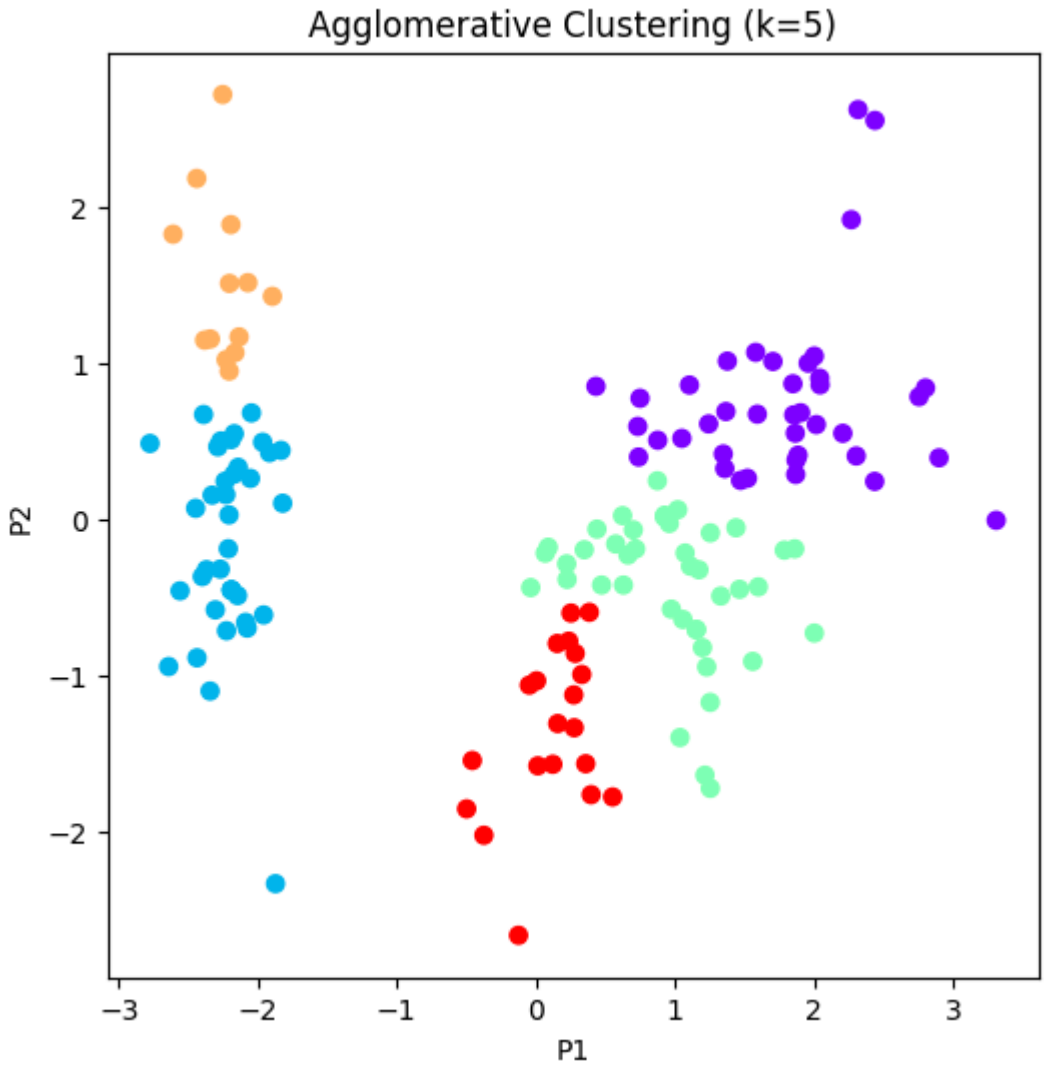


In [13]:
```
for k in range(3,6):
    models = AgglomerativeClustering(n_clusters=k)
    labels = models.fit_predict(X_principal)

    plt.figure(figsize=(6, 6))
    plt.scatter(X_principal['P1'], X_principal['P2'], c=labels, cmap='rainbow')
    plt.title(f'Agglomerative Clustering (k={k})')
    plt.xlabel('P1')
    plt.ylabel('P2')
    plt.show()
```


Agglomerative Clustering (k=3)
P2
P1
2
1
0
−1
−2
−3
−2
−1
0
1
2
3


Agglomerative Clustering (k=4)
P2
P1
2
1
0
−1
−2
−3
−2
−1
0
1
2
3



In [17]:
```
silhouetteScore = []
for k in range(2,6):
    models = AgglomerativeClustering(n_clusters=k)
    labels = models.fit_predict(X_principal)
    score = silhouette_score(X_principal,labels)
    silhouetteScore.append(score)
    print(f"silhouette Score for k={k}: {score:.4f}")

plt.figure(figsize=(8, 5))
plt.bar(range(2,6), silhouetteScore, color='skyblue')
plt.xlabel('Number of clusters (k)', fontsize=14)
plt.ylabel('Silhouette Score', fontsize=14)
plt.title('Silhouette Scores for Different k', fontsize=16)
plt.show()
```

```
silhouette Score for k=2: 0.6130
silhouette Score for k=3: 0.5103
silhouette Score for k=4: 0.4603
silhouette Score for k=5: 0.4116
```

phần 2

3. Bài tập: Thực hành phân cụm phân cấp trên dữ liệu shopping data (shopping_data.csv)

- Yêu cầu:
  - 3.1. Hiểu và tiền xử lý dữ liệu
  - 3.2. Chỉ sử dụng 2 thuộc tính 'Annual Income k$' và 'Spending Score' để trực quan hóa phân bố các điểm dữ liệu trên biểu đồ phân tán (scatter plot)
  - 3.3. Trực quan hóa quá trình phân cấp cụm dữ liệu bằng dendrogram.
  - 3.4. Xây dựng mô hình phân cụm phân cấp AgglomerativeClustering với n_clusters = 4. Vẽ biểu đồ trực quan hóa kết quả phân cụm.
  - 3.5. Đánh giá chất lượng phân cụm bằng chỉ số silhouette_score.

In [18]:
```
import pandas as pd
import matplotlib.pyplot as plt
import scipy.cluster.hierarchy as shc
from sklearn.cluster import AgglomerativeClustering
from sklearn.metrics import silhouette_score
import os
```

In [19]:
```
path = r"Z:\DOCUMENTS\KPDL\DataMining-Homework\Lab6-Phan-cum\shopping_data.csv"
```

In [20]:
```
df = pd.read_csv(path)
print("---Thông tin bộ dự liệu---")
print(df.info())
print("\n---5 dong đầu tiên ---")
print(df.head())
```

```
---Thông tin bộ dự liệu---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Genre                   200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
None

---5 dong đầu tiên ---
   CustomerID   Genre  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```

In [21]:

```python
# Cột 3: Annual Income (k$), Cột 4: Spending Score (1-100)
data = df.iloc[:, [3, 4]].values
data
```

Out[21]:
```
array([[ 15,  39],
       [ 15,  81],
       [ 16,   6],
       [ 16,  77],
       [ 17,  40],
       [ 17,  76],
       [ 18,   6],
       [ 18,  94],
       [ 19,   3],
       [ 19,  72],
       [ 19,  14],
       [ 19,  99],
       [ 20,  15],
       [ 20,  77],
       [ 20,  13],
       [ 20,  79],
       [ 21,  35],
       [ 21,  66],
       [ 23,  29],
       [ 23,  98],
       [ 24,  35],
       [ 24,  73],
       [ 25,   5],
       [ 25,  73],
       [ 28,  14],
       [ 28,  82],
       [ 28,  32],
       [ 28,  61],
       [ 29,  31],
       [ 29,  87],
       [ 30,   4],
       [ 30,  73],
       [ 33,   4],
       [ 33,  92],
       [ 33,  14],
       [ 33,  81],
       [ 34,  17],
       [ 34,  73],
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       [ 37,  75],
       [ 38,  35],
       [ 38,  92],
       [ 39,  36],
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       [ 40,  42],
       [ 40,  42],
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       [ 42,  60],
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       [ 43,  60],
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       [ 43,  41],
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       [ 46,  46],
```

```
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[ 64,  46],
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[ 65,  59],
[ 67,  43],
[ 67,  57],
```

```
[ 67,  56],
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[ 71,  75],
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[ 73,  88],
[ 73,   7],
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[ 77,  12],
[ 77,  97],
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[ 86,  95],
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[ 87,  63],
[ 87,  13],
[ 87,  75],
[ 87,  10],
[ 87,  92],
[ 88,  13],
[ 88,  86],
[ 88,  15],
[ 88,  69],
[ 93,  14],
[ 93,  90],
```
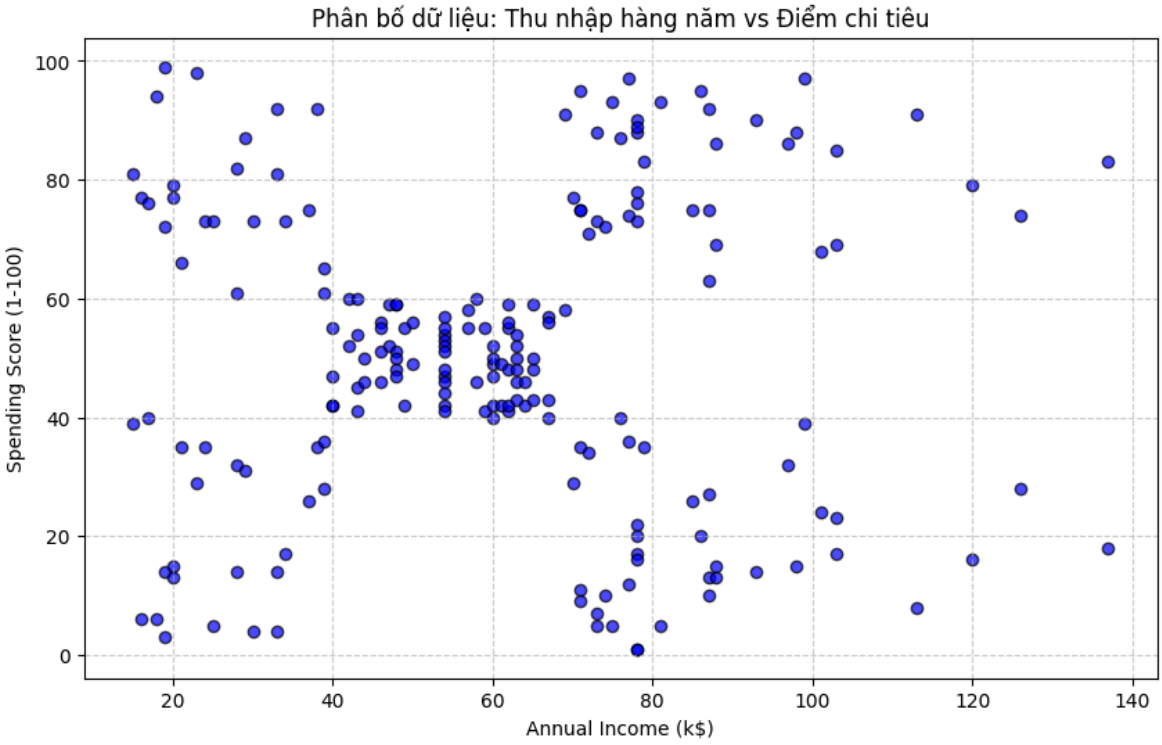
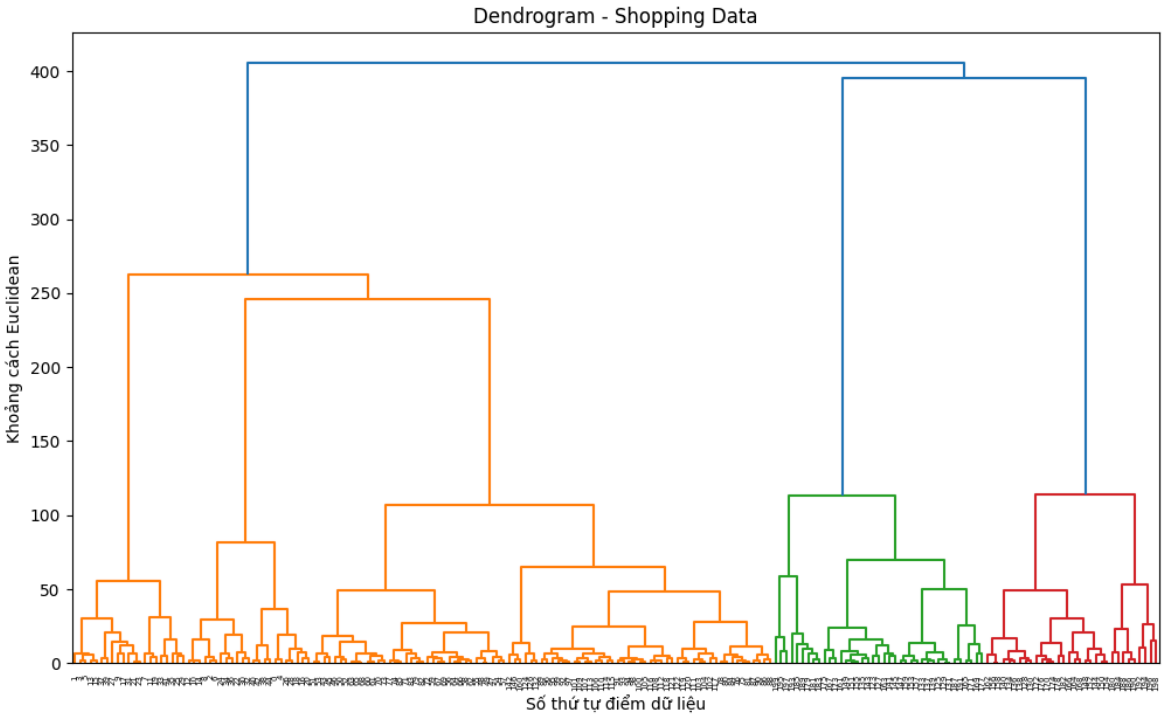

```
       [ 97,  32],
       [ 97,  86],
       [ 98,  15],
       [ 98,  88],
       [ 99,  39],
       [ 99,  97],
       [101,  24],
       [101,  68],
       [103,  17],
       [103,  85],
       [103,  23],
       [103,  69],
       [113,   8],
       [113,  91],
       [120,  16],
       [120,  79],
       [126,  28],
       [126,  74],
       [137,  18],
       [137,  83]])
```

In [22]:

```
plt.figure(figsize=(10, 6))
plt.scatter(data[:, 0], data[:, 1], color='blue', edgecolors='k', alpha=0.7)
plt.title("Phân bố dữ liệu: Thu nhập hàng năm vs Điểm chi tiêu")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```

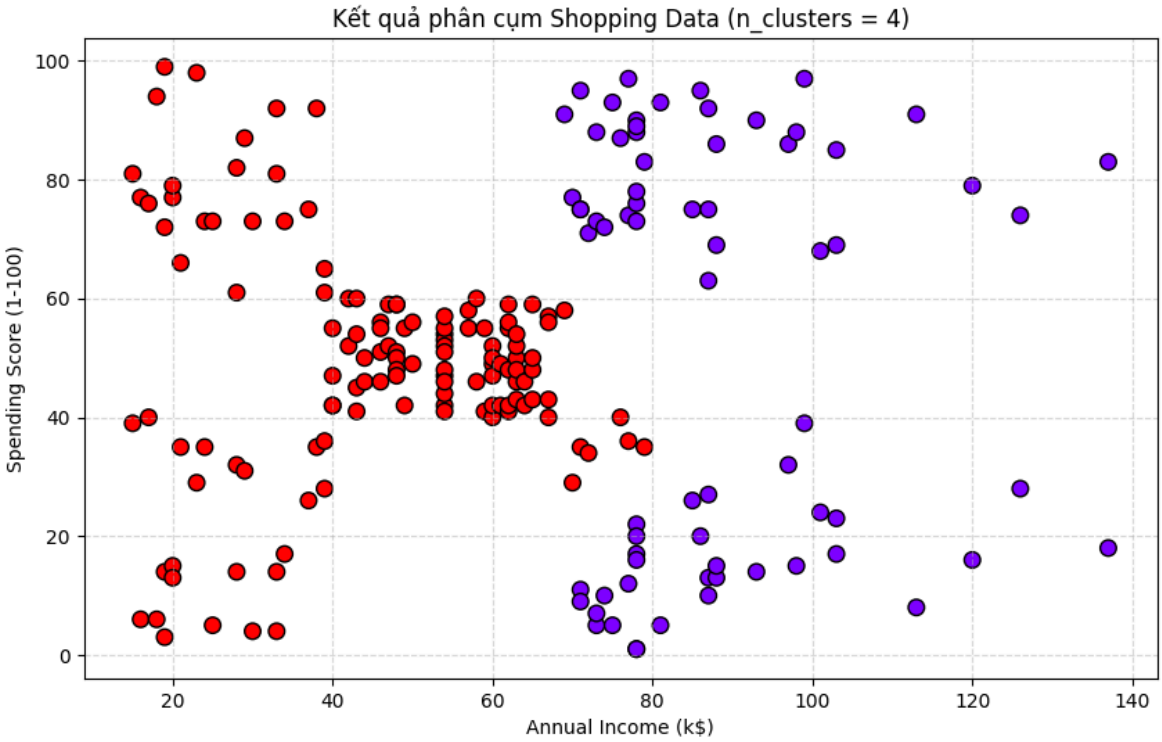


In [23]:
```
# 3.3. Trực quan hóa quá trình phân cấp bằng Dendrogram
plt.figure(figsize=(12, 7))
plt.title("Dendrogram - Shopping Data")
# Sử dụng phương pháp 'ward' để giảm thiểu phương sai trong cụm
dend = shc.dendrogram(shc.linkage(data, method='ward'))
plt.xlabel("Số thứ tự điểm dữ liệu")
plt.ylabel("Khoảng cách Euclidean")
plt.show()
```



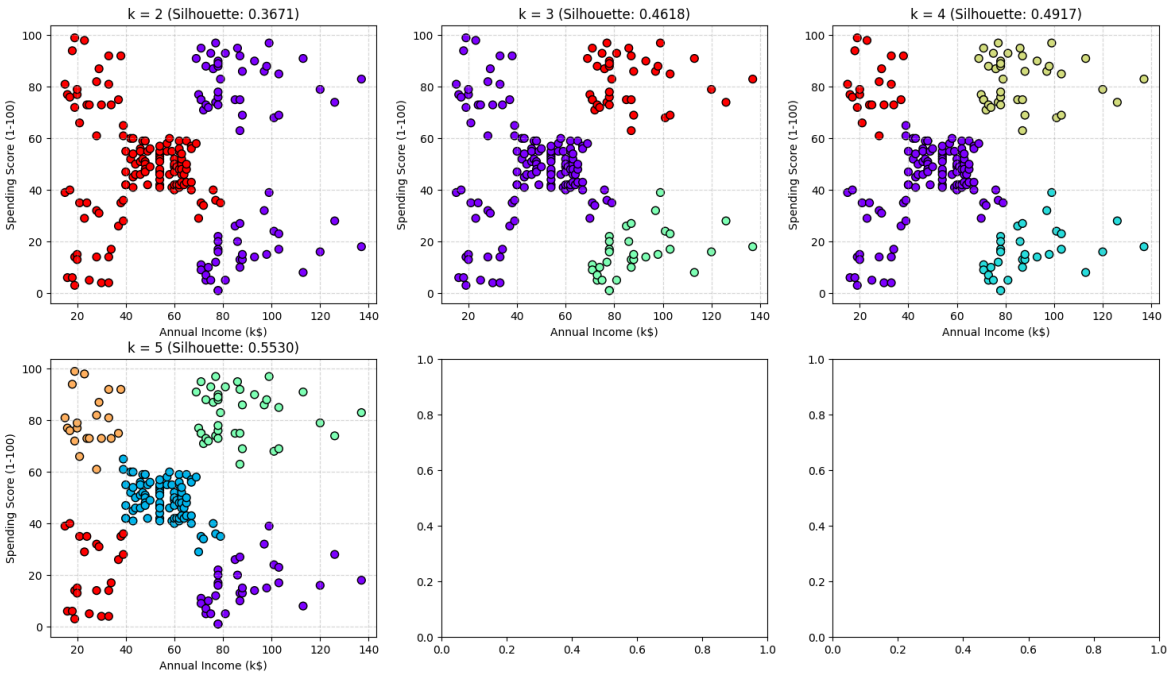
In [ ]:
```
# 3.4. Xây dựng mô hình AgglomerativeClustering với n_clusters = 5 ---
model = AgglomerativeClustering(n_clusters=2, metric='euclidean', linkage='ward'
labels = model.fit_predict(data)
```

In [27]:
```
# Trực quan hóa kết quả phân cụm
plt.figure(figsize=(10, 6))
plt.scatter(data[:, 0], data[:, 1], c=labels, cmap='rainbow', edgecolors='k', s=
plt.title("Kết quả phân cụm Shopping Data (n_clusters = 4)")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```



In [29]:
```
fig, axes = plt.subplots(2, 3, figsize=(18, 10))
axes = axes.flatten()
for i,k in enumerate(range(2,6)):
    models = AgglomerativeClustering(n_clusters=k)
    labels = models.fit_predict(data)
    score = silhouette_score(data, labels)
    print(f"Số cụm k = {k}: Silhouette Score = {score:.4f}")
    # Trực quan hóa lên subplot tương ứng
    scatter = axes[i].scatter(data[:, 0], data[:, 1], c=labels, cmap='rainbow',
    axes[i].set_title(f"k = {k} (Silhouette: {score:.4f})")
    axes[i].set_xlabel("Annual Income (k$)")
    axes[i].set_ylabel("Spending Score (1-100)")
    axes[i].grid(True, linestyle='--', alpha=0.5)
```

```
Số cụm k = 2: Silhouette Score = 0.3671
Số cụm k = 3: Silhouette Score = 0.4618
Số cụm k = 4: Silhouette Score = 0.4917
Số cụm k = 5: Silhouette Score = 0.5530
```


k = 2 (Silhouette: 0.3671)
k = 3 (Silhouette: 0.4618)
k = 4 (Silhouette: 0.4917)
k = 5 (Silhouette: 0.5530)
Spending Score (1-100)
Annual Income (k$)